

Apriori Analysis as a showcase of the services offered by an EdTech firm

A Proposal report for the BDM capstone Project

Submitted by

Name: Vedanshi Vaghela

Roll number: 23f2003037



IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai Tamil Nadu, India, 600036

Contents

1. Executive Summary and Title
2. Organisation Background
3. Problem Statement (Listed as objectives)
4. Background of the Problem
5. Problem Solving Approach
6. Expected Timeline
7. Expected Outcome

Declaration Statement

I am working on a Project Title Apriori Analysis as a showcase of the services offered by an EdTech firm. I extend my appreciation to **SproutSong**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate: 

Name: Vedanshi Vaghela

Date: 12/05/2025

1.Executive Summary and Title

- SproutSong is an AI-powered education technology platform that empowers students, universities, schools, and institutions through personalized counselling, smart analytics, and career guidance.
- The firm is seeking a concise yet meaningful sample data analysis that can be presented to potential partner institutions. This will help demonstrate the value of its offerings and support its partnership outreach, especially as it is currently in the early stages of its launch.
- My approach involves performing exploratory data analysis (EDA) on the sample student dataset provided by the firm, followed by clustering to identify meaningful patterns or groupings. Based on the insights derived, I will offer data-driven suggestions for potential improvements in the current program structure, if applicable. These insights will be showcased on the firm's website as part of their portal to demonstrate the platform's analytical capabilities.

The project focuses on a B2C and B2B organized educational startup based in Gandhinagar, Gujarat, that connects parents, students, and institutions to support career development and learning. As a newly launched firm, it faces challenges in forming partnerships due to limited visibility and experience in outreach, marketing, and stakeholder engagement, making it difficult to effectively convey its value proposition.

The issues will be addressed through a combination of exploratory and statistical data analysis techniques using tools such as OpenRefine, Google Sheets, and Python. Key approaches will include:

- **Exploratory Data Analysis (EDA):** To identify trends, outliers, and patterns within student performance and engagement data.
- **Clustering Techniques:** Such as K-Means or Hierarchical Clustering to group students based on performance or behavior, helping to tailor program strategies.
- **Statistical Tests:** Including ANOVA (Analysis of Variance) to compare performance across different groups or time periods.

These methods aim to extract actionable insights that can inform program improvements and enhance the firm's ability to build stronger partnerships.

The expected outcome helps the firm increase their audience engagement and pave way for new partnerships.

2.Organization Background

SproutSong is a for-profit EdTech startup founded in 2024 by Mr. Keyur Rathod and based in Gandhinagar, Gujarat. It offers a comprehensive, board-agnostic educational ecosystem supporting students from Standards 5 to 12 across GSEB, CBSE, and ICSE. Integrating AI, machine learning, and data-driven insights, SproutSong promotes holistic student development—from academic progress and stream selection to motivation and institutional support. The platform emphasizes conceptual learning and continuous improvement in the students' development through interactive content and real-world applications, fostering curiosity and critical thinking. SproutSong delivers tailored solutions for students, parents, teachers, and schools, encouraging collaborative growth. In addition to academic tools, it hosts workshops and programs focused on motivation, stream selection, and skill-building, empowering students to discover and pursue their potential. With smart dashboards and reporting features, SproutSong brings clarity and direction to the educational journey, serving as a reliable guide beyond exams and grades. Its mission is to enhance learning experiences and drive meaningful educational outcomes across diverse learning environments.

3.Problem Statement (Listed as objectives)

1. **Explore Student Performance Data:** Use EDA to uncover trends, patterns, and outliers in student engagement and academic data.
2. **Segment Student Profiles:** Apply clustering techniques (e.g., K-Means, Hierarchical Clustering) to group students based on performance, enabling more personalized program strategies.
3. **Evaluate Program Effectiveness:** Use statistical tools such as ANOVA to compare student outcomes across different groups or time periods, and give a performance distribution display on the dash-board.

4.Background of the Problem

The firm is currently in the early stages of its launch and is facing several challenges in establishing meaningful partnerships with both educational institutions and parents. As a new entrant in the education and career

development sector, it has yet to build strong brand recognition or establish trust among its target audience. One of the primary hurdles is the limited awareness of the firm's existence and offerings. Since the company is still developing its presence in the market, many potential stakeholders, including schools, teachers, and parents, may not be fully informed about the platform's capabilities or the value it brings to the educational ecosystem.

Some of the major cause of problems faced by the firm are:

1. Lack of Brand Awareness, since they're new in the market with minimal visibility, target users (parents, institutions) may not trust or recognize the brand.
2. Challenges in Building Partnerships i.e schools and parents may be hesitant to collaborate with a new, unproven entity. Lack of case studies or prior success stories makes partnership pitches harder.
3. Unclear Market Positioning i.e being board-agnostic is a strength, but it may confuse audiences about specialization.
4. Lack of Structured Feedback Loops, Without institutional or parent feedback or partnerships/connections likewise, continuous improvement becomes hard.

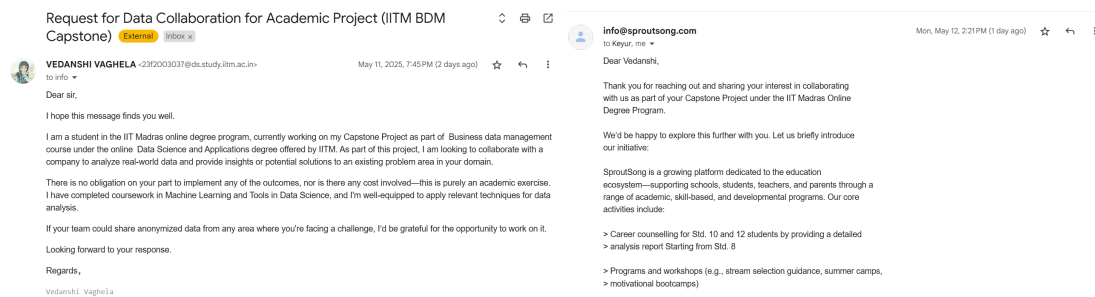
While the platform has a strong vision and model, it needs support in building market presence and conveying its value. This project aims to use data analysis and strategic insights to enhance stakeholder engagement, improve offerings, and help the firm establish long-term partnerships and expand its outreach.

5.Problem Solving Approach

To address the challenges faced by the firm, a structured approach involving a blend of exploratory and statistical data analysis techniques will be adopted which they can present to any educational institution or parents alike to promote the firm. The aim is to generate actionable insights that can enhance audience engagement, improve existing programs, and support the formation of strategic partnerships. The analysis will be carried out using tools such as **OpenRefine**, **Google Sheets**, and **Python**, chosen for their versatility and effectiveness in handling, cleaning, and analyzing structured data.

The firm has asked for some time before providing me with the concerned data, as a proof i have the following screenshots of my approach to the company, although a meet has not yet been conducted, and all the conversations have taken place via email, the

thread can be viewed -  proof of approach , some photo evidence is also attested below



The first step will involve **Exploratory Data Analysis (EDA)** to examine student performance and engagement data. EDA will help uncover trends, identify outliers, and detect underlying patterns. This step is crucial for gaining an initial understanding of the dataset and determining initial patterns and distributions in a student's overall grasp of a given subject/topic. Visualizations such as histograms, box plots, and scatter plots will be used to present these patterns clearly.

Following EDA, **clustering techniques** such as **K-Means** or **Hierarchical Clustering** will be applied to segment students into distinct groups based on similarities in performance in a given course/subject. These clusters can reveal different learning profiles, which can then inform the development of tailored strategies for each group which would ultimately benefit the students, their parents or the educational institute. For example, one cluster might consist of high performers who could benefit from advanced content, while another might include students needing additional support. Such segmentation allows for more focused interventions and program planning.

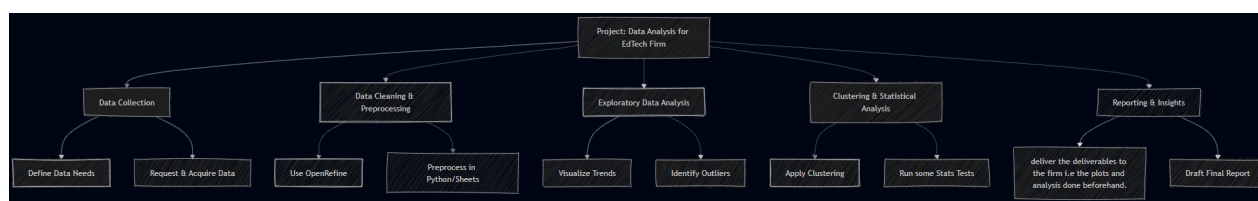
In addition to clustering, this project will employ **statistical tests**, particularly **ANOVA (Analysis of Variance)** or **finding the underlying distribution of students overall grasp and growth/downfall over the academic year in the student's performance** to assess whether there are significant differences in performance across various groups, such as students from different grades or assessment periods from which an educational institution could benefit. This analysis helps determine which factors could have a measurable impact on student outcomes and where improvements might be most needed.

The combined use of these methods is expected to provide a comprehensive view of how students are engaging with the platform and where the educational programs might be optimized. More importantly, it will equip the firm with data-backed insights to support

their outreach to parents and institutions, reinforcing credibility and demonstrating value. Ultimately, the project aims to pave the way for increased engagement, improved academic support, and the establishment of new institutional partnerships, positioning the firm for long-term growth and impact in the education sector.

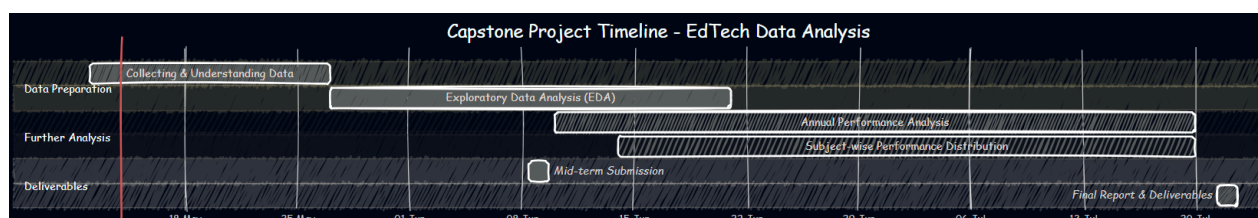
6.Expected Timeline

1. Work Breakdown Structure:



This flowchart outlines the key stages of the EdTech data analysis project, from data collection to reporting. Besides EDA, the major tasks include **Annual Performance Analysis** and **Subject-wise Performance Distribution** as part of the statistical analysis phase.

2. Gantt chart



The Gantt chart outlines the capstone project timeline, starting with data collection and EDA, followed by in-depth analysis and final deliverables over a span of about 2.5 months. The two key tasks beyond EDA are **Annual Performance Analysis** and **Subject-wise Performance Distribution**.

7.Expected Outcome

The expected outcome of this project includes deliverables aimed at offering both immediate impact and long-term value to the firm. The analysis will produce key visualizations and performance charts that can be showcased on the company's website or used in presentations to highlight trends and insights. These will support clearer communication and strengthen credibility with potential partners and stakeholders.

The project will conclude with a set of tailored recommendations drawn from the analysis, aimed at helping the firm refine its outreach strategy, build more targeted programs, and foster stronger partnerships with educational institutions and parents. These efforts will support both internal improvements and broader organizational growth.

TA comments:

The page count is not within the limits (the proposal report should be limited to 4-5 pages. It can be of 7-8 pages including content page, title page, declaration statement etc. as per rubrics). There are no page numbers in contents page. The font style, spacing and text alignment should be uniform throughout the report. You should have proper numbering for each section and subsection for better readability (for example 5 for main section, 5.1 for subsection, 5.1.1 for sub-sub-section etc). Section1: This section is not okay as per rubrics. The title should just be "1 Executive Summary". This section should be written in paragraphs. You should avoid bullet points and sub-sections within this section. It should have a brief summary of all the subsequent sections in the report in some 2-3 paragraphs and the same will be elaborated further. You should have briefly mentioned about what kind of data, solution approach, tools (python/excel etc.) that you are planning to use, expected outcomes of the project in this section. Section2: This section has to be better. Some additional information on products, number of employees, annual revenue would have made this section stronger. Section3: Typically, business problems are about increasing profits/sales/revenue or decreasing wastage/loss etc. There is relevant content in this section, but problem has to be properly formulated as a business problem by linking to a business objective. You need to find a way to link student performance objective to a business objective here. Section4: This section somewhat looks okay. Section5: Screenshots are not relevant here. The rest of the section has relevant content but should be organized properly as per rubrics. In this section you should have sub-sections about what kind of data will be collected for analysis and its source, tools used for analysis (excel or python etc) and the solving approach relevant to each of the problem statements (for example regression is a solution method for a business problem of demand forecasting). The expectation from the BDM project is having at least one specific quantitative data analysis-based solution approach for each of the problem statement considered. Rest of the report broadly looks okay. There is relevant content in your report. You should just modify the report as per the review comments (particularly section1, 5) and resubmit in the next cycle to meet BDM project rubrics.